

EVIDENCE DOSSIER

Kernellum Evidence Dossier

A claim-to-artifact record for the compiler, routed-search, SIMILARITY, and K2 evidence

RESEARCH LANE

EVIDENCE CONTROL / PUBLICATION BUNDLE

REPORT ID

KRN / EVD-001

DATE

21 September 2026

STATUS

External research evidence dossier

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PURPOSE

Evidence that scales with the claim

This dossier is the external claim-control layer for Kernellum Research. It packages the current technical reports, machine-readable evidence, implementation records, and K2 validation handoff into one coherent research-lab publication bundle.

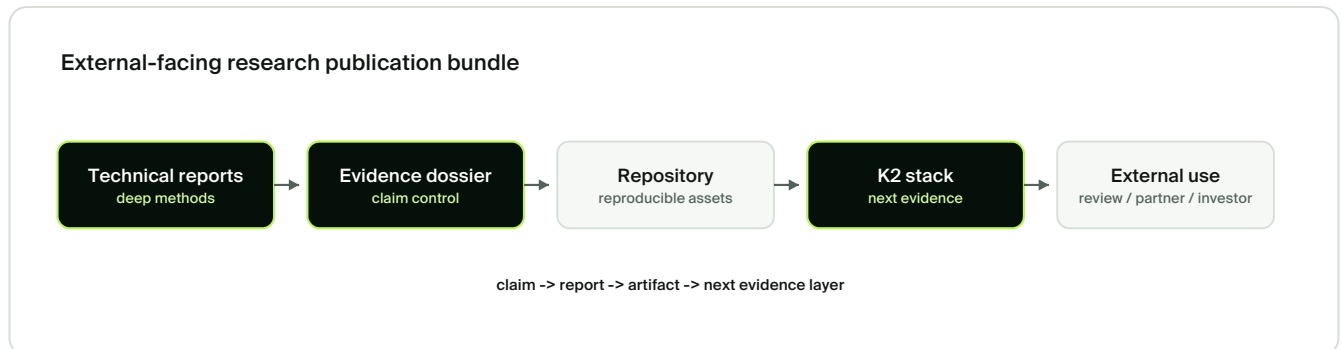


Figure 1. External-facing publication bundle. Technical reports, the evidence dossier, repository artifacts, and the K2 handoff form one connected evidence stack.



Program at a glance

Kernellum now has a reproducible model-to-RTL compiler path, final-route architecture ranking, a bounded closed-loop search result, a causal routed timing law, a route-robust two-candidate selection policy, and a frozen K2 board-validation stack.

01 / PORTFOLIO

Evidence maturity

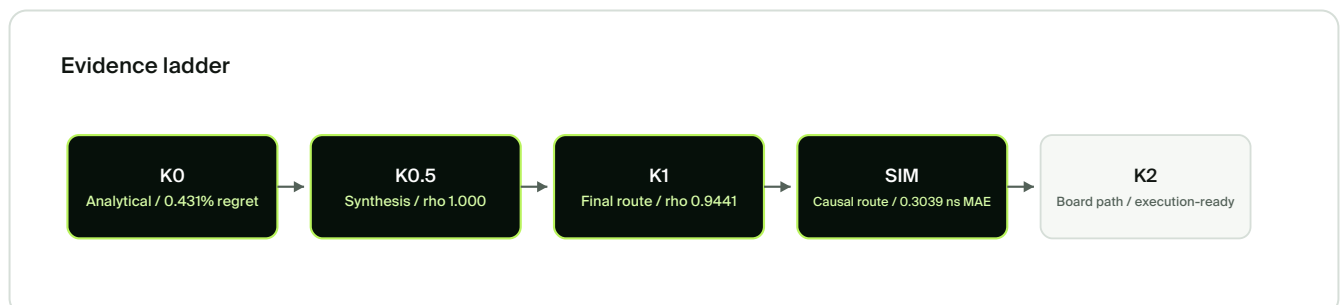


Figure 2. Program maturity from analytical search through final-route evidence and the K2 physical-validation handoff.

Evidence maturity matrix

	Software	RTL	Synth	Route	Board
Compiler	PASS	PASS	PASS		
K0	PASS				
K0.5	PASS	PASS	PASS		
K1	PASS	PASS	PASS	PASS	
K1 loop	PASS	PASS	PASS	PASS	
SIM law	PASS	PASS	PASS	PASS	
SIM portfolio	PASS	PASS	PASS	PASS	
K2 stack	PASS	PASS	PASS	PASS	NEXT

Figure 3. Evidence matrix by program. The current public record is strongest at final route; K2 is the structured next layer.

The matrix keeps each public statement attached to the strongest evidence actually present in the repository, making the dossier useful for technical review, partner diligence, and investor-facing research communication.

02 / STATUS

Current evidence snapshot

Program	Research question	Current result
Compiler	Can the model-to-RTL path be reproduced?	96.44% INT8; 99.78% agreement; 450/450 cycle exact; 32/32 RTL golden cases
K0	Can budgeted search recover analytical optima?	0.431% regret @256 evals; 0.069% @512
K0.5	Do resource rankings survive synthesis?	DSP and BRAM rank Spearman = 1.000
K1	Do rankings survive final route?	9/9 routed; rank Spearman 0.9441; winner regret 0.5828%
K1 closed loop	Can routed observations guide a fixed budget?	8/8 new routes; 6.475% MAPE; 12/12 workloads improved
SIM routed law	Does topology create a scalable routed effect?	96/96 routes; held-out MAE 0.3039 ns; 13/13 gates
SIM portfolio	Can two candidates absorb route-seed variation?	72/72 routes; 36/36 deployment wins; 23.56% gain; 1.44% regret
K2 stack	Is the board-validation package execution-ready?	Three routed bitstreams, UART protocol, frozen host runner, hashes

03 / REPRODUCIBILITY

Claim-to-artifact traceability

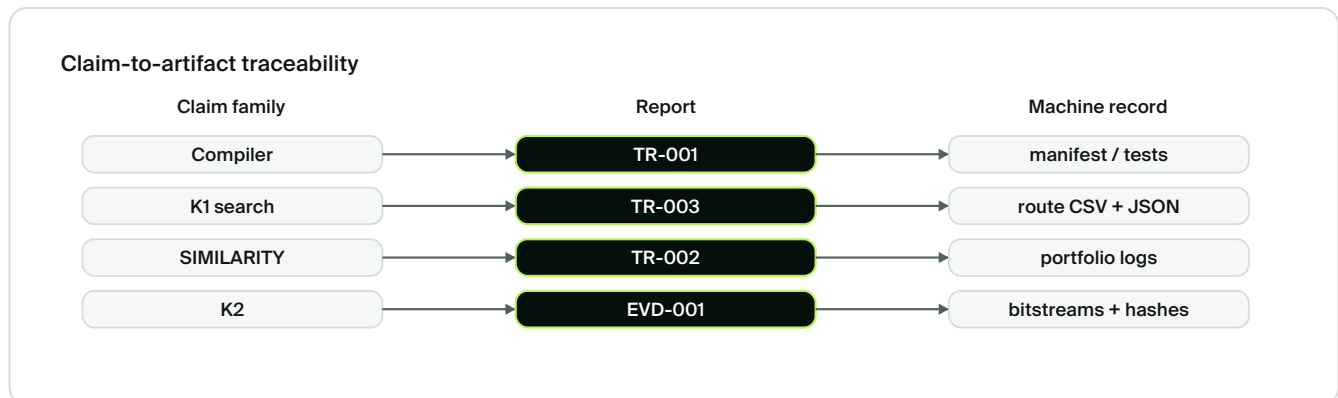


Figure 4. Public claims map through a named report to a machine-readable artifact family.

Claim family	Primary record
Compiler	artifacts/digits_int8/manifest.json / tests/test_pipeline.py
K1 ranking	results/k1_routes.csv / results/k1_validation.json
K1 closed loop	results/k1_closed_loop_routes.csv / results/k1_closed_loop_validation.json
SIM causal law	results/similarity_routed_law_combined.csv / similarity_routed_law_summary.json
SIM portfolio	results/similarity_portfolio_combined.csv / similarity_portfolio_summary.json
K2 board stack	results/k2_board_ready_manifest.json / docs/K2_BOARD_READY_REPORT.md

04 / NEXT EVIDENCE LAYER

K2 board-validation readiness

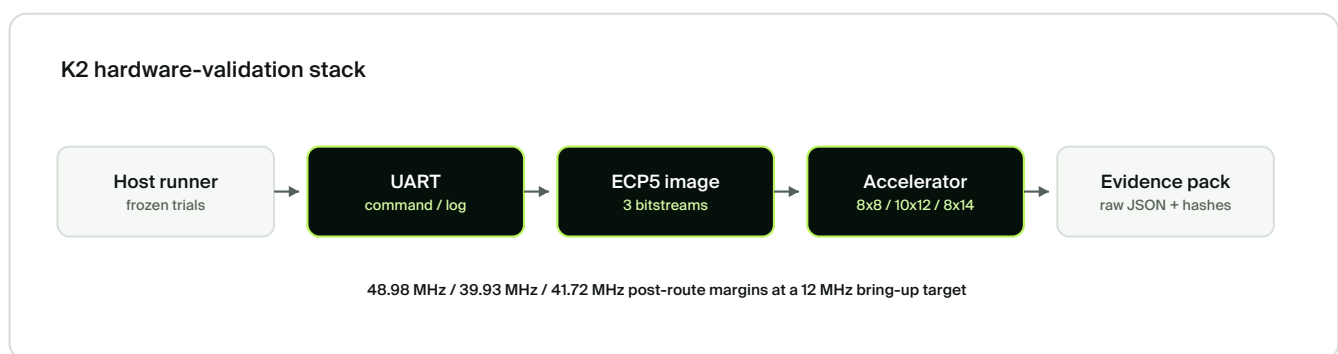


Figure 5. K2 validation stack. The board package is designed so the next release can move cleanly from routed evidence to physical timing records without changing the research-control structure.

Variant	DSP	BRAM	LUT4	FF	Post-route Fmax
8x8 K32	64	4	8,928	2,551	48.98 MHz
10x12 K32	120	6	11,898	4,439	39.93 MHz
8x14 K64	112	6	15,055	4,188	41.72 MHz

Near-term value inflection

The next publication layer is a physical-board evidence package: frozen bitstreams, deterministic UART trials, raw logs, timing summaries, and traceable hashes. This extends the current routed record rather than replacing it.

05 / EXTERNAL LANGUAGE

Public research statements

The following statements are supported directly by the current report and artifact set:

- Kernellum has a reproducible model-to-RTL compiler demonstration with exact cycle-model agreement and held-out RTL simulation.
- Analytical architecture rankings remain strongly correlated with final-route rankings in the frozen K1 ECP5 design family.
- Sparse final-route observations can guide a fixed physical-design budget more effectively than the equal-budget random control in the tested K1 space.
- Operand-distribution topology produces a measurable size-dependent routed timing effect in the tested ECP5 INT8 MAC-fabric family.
- A frozen differential law predicts unseen rectangular routing tax with **0.3039 ns** MAE.
- A two-candidate local portfolio beats the best broadcast-only architecture in **36/36** unseen deployment cases with **23.56%** mean improvement and **1.44%** mean routed-oracle regret.
- The K2 hardware-validation stack is packaged for execution under the frozen bring-up protocol.

Investor + research-lab framing

Kernellum is presented as a research system rather than a collection of isolated experiments: the site states concise claims, the reports provide methods and diagrams, the dossier controls claim strength, and the repository carries reproducible machine records.

Primary reproduction map

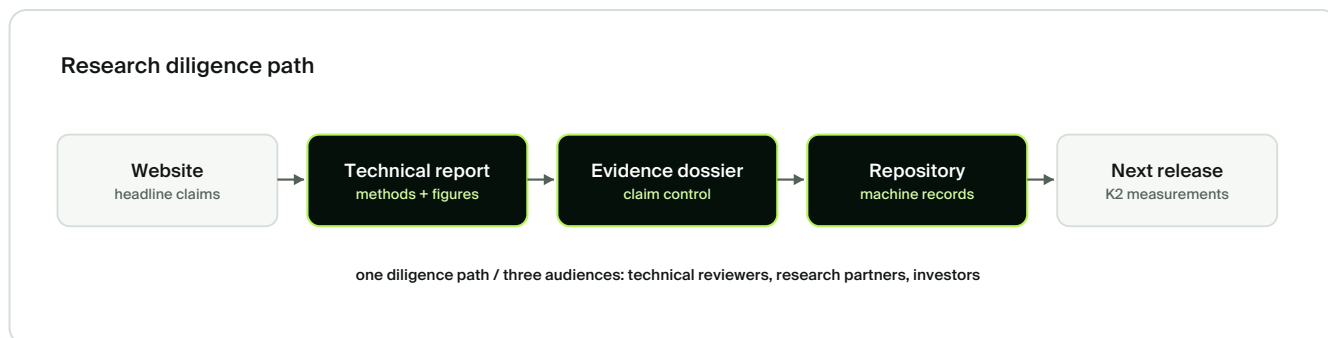


Figure 6. Research diligence path. The same evidence stack can be traversed from concise website claim to report, dossier, repository record, and the next K2 physical-validation release.

Record	Path
Experiment index	docs/EXPERIMENT_LEDGER.md
K1 correction	docs/K1_FINAL_ROUTE_CORRECTION_REPORT.md
SIM causal report	docs/SIMILARITY_ROUTED_LAW_REPORT.md
SIM deployment study	docs/SIMILARITY_DEPLOYMENT_CONFIRMATION_REPORT.md
SIM portfolio confirmation	docs/SIMILARITY_PORTFOLIO_CONFIRMATION_REPORT.md
K1 closed loop	docs/K1_CLOSED_LOOP_REPORT.md
K2 board-ready	docs/K2_BOARD_READY_REPORT.md
Public repository	github.com/sushxnthd/kernellum

Document record

KRN / EVD-001. Kernellum Research. 21 September 2026. This dossier is designed for technical reviewers, research partners, and diligence workflows that need a concise path from headline claim to reproducible artifact.

How to use the bundle

Technical review starts with the report methods and figures; diligence checks the evidence dossier and machine records; partnership and investment conversations can use the same traceability path without reducing the work to marketing claims.